

VR Education on Historic Lunar Roving Missions

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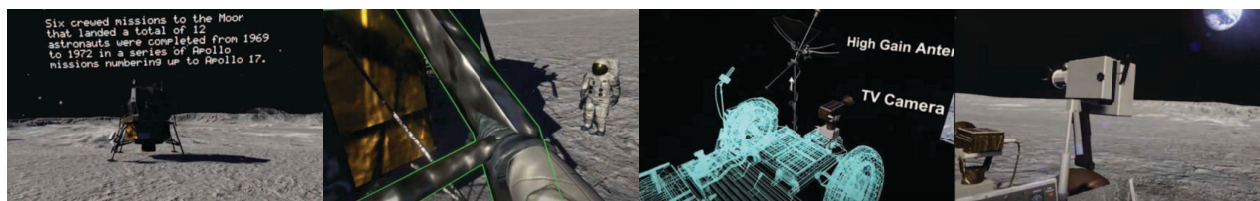


Figure 1: The educational VR game for teaching historic Apollo lunar roving missions.

ABSTRACT

This work presents the design and evaluation of an educational VR game to teach historic Apollo lunar roving missions. The game consists of three gameplay scenes, and each scene aims to convey one type of knowledge. The game adheres with the learning objectives, presents the contents, and conveys knowledge in both active and passive interaction modes. We conducted a user study focusing on the understanding of the influence of different interaction modes on learning outcomes and learning engagement.

Index Terms: Educational VR games, serious VR games, informal learning, historic event teaching

1 INTRODUCTION

VR technologies have shown pedagogical promise in the presentation of historic facts, allowing learners to see and interact with what had happened [1]. For the individuals who have not engaged with a specific historic event, the VR presentation of the event with a mix of interactive graphical and audio media strives for new ways to depict knowledge of the past and improve their desire to learn about it. In accordance with the *situated learning theory* [2], learners benefit from being engaged with the contexts of knowledge through a teaching placement where they are immersed within an environment that replicates an actual one. Interacting with a learning environment in VR has the potential to create an experience that improves the amount of information they recall [6] and enhance spatial understanding [5]. The design and evaluation of educational VR games are often context-dependent and should adhere with the goal of instructional design theories [4], which is to align the relationship amongst *learning objectives*, *learning activities*, and *learning*

assessment.

The study case in this work is the historic Apollo lunar roving missions, for which we came up with a new game design based on the learning objectives exhibited in NASA education programs and museums. The learning assessment is developed during the design process to make alignment with the learning objectives and learning activities. The learning activities in VR contains *active* and *passive* interaction modes.

2 THE DESIGN PROCESS

Designing an educational VR game is a complex process requiring the understanding of specific domain it applies to. The design of this educational VR game extended the gamified design presented by Peng et al. [7] and enriched the VR content presented by Cao et al. [1]. Particularly, the effort in this work focused on education that delivers learning objectives and enhances the understanding of the difficulties of this historic mission and its technical uniqueness.

2.1 Learning Objectives

The story structure in our game is composed of three scenes noted as *Landing*, *Inspecting*, and *Driving*, as the storyboard shown in Figure 2 indicates. The scenes represent different thematic motifs of the lunar roving mission, and each scene aims to convey one type of knowledge, including *Spatial knowledge*, *Factual knowledge*, and *Operating knowledge*.

- The spatial knowledge acts as a skill to find relative positions, recall/describe locations, sizes, distance, etc.
- The factual knowledge is associated with actual details of scientific facts.
- The operating knowledge to operate unfamiliar products is an ability to translate rules and procedures into performance.

In the Scene *Landing*, the game starts inside of the lunar landing module. The learner needs to exit the hatch, climb down the ladder, and see an animated deployment process that detaches the rover from the landing module.

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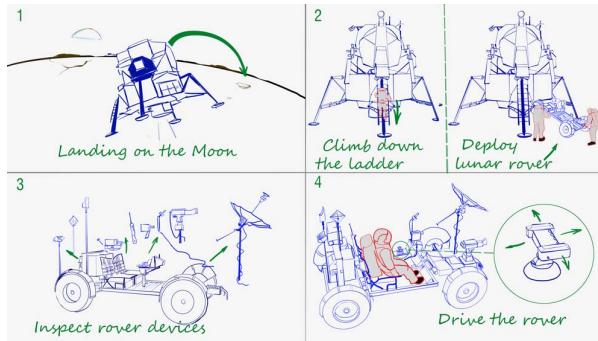


Figure 2: The storyboard of this lunar roving game for the introduction scene and three gameplay scenes.



Figure 3: (a) The photo taken during the user study. (b) The screenshot of the VR content that the participant was inspecting.

In the Scene *Inspecting*, the learner inspects a list of devices of the rover and learns about their usages. Animations have been considered as an effective medium to express factual knowledge [3].

In the scene *Driving*, the learner is able to experience sensations of real-world side of rover driving and relevant practical issues.

2.2 Learning Activities

In this work, we examined two interaction modes, *active* and *passive*, to find out effects towards learning experiences in VR and learners' acquisition of different types of knowledge. In the active mode, the learner controls his or her movements, determines the gameplay speed, and decides the learning order when choices are available. On the contrary, the passive mode conducts the learning activities as a viewing observation where movements and events are programmed into the story structure.

3 USER STUDY

We developed this game using Unity Game engine. We recruited 26 participants and randomly assigned active mode or passive mode for participants to play the game. Figure 3 is a photo of the user study. We used the *between subjects* study for the active and passive interaction modes, where each mode of the game was played by 13 participants. The entire study took about 45 minutes including 20 minutes of gameplay and 25 minutes of answering questionnaires.

4 RESULTS

We used the Friedman test to analyze the answers in the questionnaires (discrete data). All statistical results were based on the significance value of $\alpha = .05$.

4.1 Learning Outcome Evaluation

In observing the results in Learning Evaluation Questionnaire (LEQ), although no statistical significance between the active and passive modes, the active mode is likely to be more effectively to

convey spatial knowledge, and the passive mode is likely to be more effectively to convey factual knowledge.

We asked the participants to give a rating after the study about the learning effectiveness of the interaction mode they used, with a five-point scale, where 1 means ineffective and 5 means very effective. The active mode was rated higher than the passive mode for all scenes, where for the scene *Landing* the rates for active and passive modes were 4.69 and 4.23, for the scene *Inspecting* they were 3.85 and 3.69, and for the scene *Driving* they were 4.69 and 4.38, respectively. The participants also gave a rating about the quality of the content of all scenes with a five-point scale, where 1 means bad and 5 means excellent. The passive mode was rated higher than the active mode, where the rates were 4.23 (passive) and 3.69 (active), but we didn't find significant differences in the ratings between the passive and active modes.

4.2 User Experience Evaluation

We grouped the 26 items of the User Experience Questionnaire (UEQ) into six categories: attractiveness, perspicuity, efficiency, dependability, stimulation, and novelty [8]. The mean attractiveness rating of the active mode (Mean = 6.29, SD = 0.55) is higher than the passive mode (Mean = 5.94, SD = 0.103). For other categories, no significant difference was found between the two modes.

5 CONCLUSION AND FUTURE WORK

In this work, we have designed and developed an educational VR game that represents the historic Apollo lunar roving missions performed by NASA astronauts during the late 1960s and early 1970s. The game enables a series of informal learning experiences to learn about spatial, factual, and operating knowledge through the view from the perspective of astronauts who lived through this historic mission. We conducted a user study to learn the difference between active and passive modes. In the future, we plan to add a gesture recognition method to enable the freehand interaction with the VR content in the active mode.

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